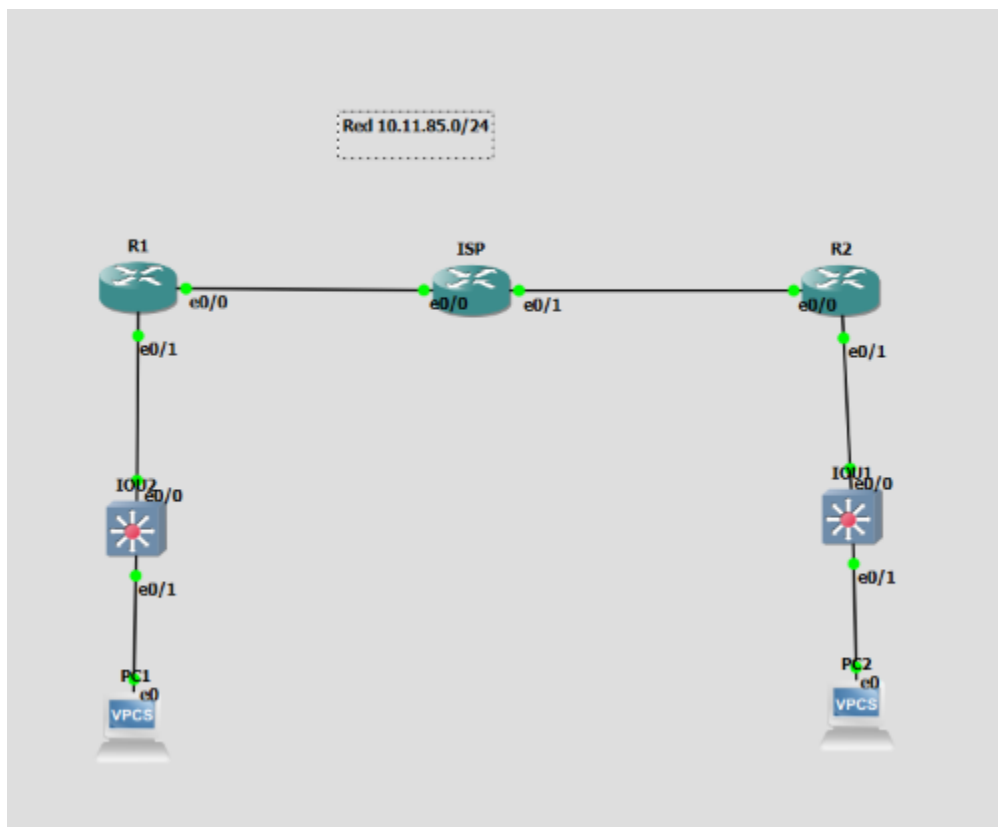


. Objetivo

Implementar VPN Site-to-Site con IPSec IKEv2 policy-based. IKEv2 introduce el concepto de keyring (claves por peer) y profile (agrupa autenticación + keyring), más seguro y flexible que IKEv1. El tráfico interesante sigue siendo definido por ACL extendida. La diferencia más visible en la config es set ikev2-profile dentro del crypto map.

2. Topología



Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Descripción
R1	e0/0	10.11.85.65/26	WAN → ISP
R1	e0/1	10.11.85.1/26	LAN → SW1
ISP	e0/0	10.11.85.66/26	Hacia R1
ISP	e0/1	10.11.85.129/26	Hacia R2
R2	e0/0	10.11.85.130/26	WAN → ISP
R2	e0/1	10.11.85.193/26	LAN → SW2
PC1	e0	10.11.85.2/26	GW: 10.11.85.1
PC2	e0	10.11.85.194/26	GW: 10.11.85.193

3. Parámetros IKEv2

Parámetro	Valor
Proposal	PROP-IKEV2
Cifrado	AES-CBC-256
Integridad	SHA-256
Grupo DH	14 (2048-bit)
Keyring / PSK	KR-IKEV2 / ITLA2024

Transform Set	TS-IKEV2
Protocolo / Modo	ESP / Tunnel
Cifrado	AES-256
Integridad	HMAC-SHA-256
Crypto Map	CM-IKEV2 + set ikev2-profile PROF-IKEV2

ACL tráfico

permit ip 10.11.85.0/26 ↔ 10.11.85.192/26

4. IKEv1 vs IKEv2

Elemento	IKEv1	IKEv2
Parámetros IKE	crypto isakmp policy	crypto ikev2 proposal + policy
Clave PSK	crypto isakmp key X addr Y	crypto ikev2 keyring → peer
Profile	No existe	crypto ikev2 profile (obligatorio)
Vincular crypto map	No necesario	set ikev2-profile en crypto map / ipsec profile
Verificación SA	show crypto isakmp sa	show crypto ikev2 sa
Estado SA activa	QM_IDLE	READY
Mensajes negociación	9 (Main + Quick Mode)	4 (IKE_SA_INIT + IKE_AUTH)

5. Configuración R1

```
enable
configure terminal
hostname R1
interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.65 255.255.255.192
no shutdown
interface Ethernet0/1
description LAN hacia SW1-PC1
ip address 10.11.85.1 255.255.255.192
no shutdown
ip route 0.0.0.0 0.0.0.0 10.11.85.66
crypto ikev2 proposal PROP-IKEV2
encryption aes-cbc-256
integrity sha256
group 14

crypto ikev2 policy POL-IKEV2
```

```

proposal PROP-IKEV2
crypto ikev2 keyring KR-IKEV2
peer R2
address 10.11.85.130
pre-shared-key ITLA2024
crypto ikev2 profile PROF-IKEV2
match identity remote address 10.11.85.130 255.255.255.255
authentication remote pre-share
authentication local pre-share
keyring local KR-IKEV2
crypto ipsec transform-set TS-IKEV2 esp-aes 256 esp-sha256-hmac
mode tunnel
ip access-list extended ACL-VPN-R1
permit ip 10.11.85.0 0.0.0.63 10.11.85.192 0.0.0.63
crypto map CM-IKEV2 10 ipsec-isakmp
set peer 10.11.85.130
set transform-set TS-IKEV2
set ikev2-profile PROF-IKEV2
match address ACL-VPN-R1
interface Ethernet0/0
crypto map CM-IKEV2

end

```

6. Configuración R2

```

enable
configure terminal
hostname R2
interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.130 255.255.255.192
no shutdown
interface Ethernet0/1
description LAN hacia SW2-PC2
ip address 10.11.85.193 255.255.255.192
no shutdown
ip route 0.0.0.0 0.0.0.0 10.11.85.129
crypto ikev2 proposal PROP-IKEV2
encryption aes-cbc-256
integrity sha256

```

```
group 14
crypto ikev2 policy POL-IKEV2
proposal PROP-IKEV2
crypto ikev2 keyring KR-IKEV2
peer R1
address 10.11.85.65
pre-shared-key ITLA2024
crypto ikev2 profile PROF-IKEV2
match identity remote address 10.11.85.65 255.255.255.255
authentication remote pre-share
authentication local pre-share
keyring local KR-IKEV2

crypto ipsec transform-set TS-IKEV2 esp-aes 256 esp-sha256-hmac
mode tunnel
ip access-list extended ACL-VPN-R2
permit ip 10.11.85.192 0.0.0.63 10.11.85.0 0.0.0.63
crypto map CM-IKEV2 10 ipsec-isakmp
set peer 10.11.85.65
set transform-set TS-IKEV2
set ikev2-profile PROF-IKEV2
match address ACL-VPN-R2
interface Ethernet0/0
crypto map CM-IKEV2

end
```

7. Configuración ISP

```
enable
configure terminal
hostname ISP

interface Ethernet0/0
ip address 10.11.85.66 255.255.255.192
no shutdown

interface Ethernet0/1
ip address 10.11.85.129 255.255.255.192
no shutdown

end
```

write memory

8. Verificación

! Estado SA IKEv2 (QM_IDLE → READY en IKEv2)

show crypto ikev2 sa

```
R1#
R1#show crypto ikev2 sa
IPv4 Crypto IKEv2 SA

Tunnel-id Local Remote fvrf/ivrf Status
1 10.11.85.65/500 10.11.85.130/500 none/none READY
Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/977 sec

IPv6 Crypto IKEv2 SA

R1#
```

```
R2#
R2#show crypto ikev2 sa
IPv4 Crypto IKEv2 SA

Tunnel-id Local Remote fvrf/ivrf Status
1 10.11.85.130/500 10.11.85.65/500 none/none READY
Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/1004 sec

IPv6 Crypto IKEv2 SA

R2#
```

show crypto ikev2 sa detailed

```
R1#show crypto ikev2 sa detailed
IPv4 Crypto IKEv2 SA

Tunnel-id Local Remote fvrf/ivrf Status
1 10.11.85.65/500 10.11.85.130/500 none/none READY
Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/1023 sec
CE id: 1001, Session-id: 1
Status Description: Negotiation done
Local spi: A3CBC2A6D732C26F Remote spi: 81AC0F3DC5D372D6
Local id: 10.11.85.65
Remote id: 10.11.85.130
Local req msg id: 2 Remote req msg id: 0
Local next msg id: 2 Remote next msg id: 0
Local req queued: 2 Remote req queued: 0
Local window: 5 Remote window: 5
DPD configured for 0 seconds, retry 0
Fragmentation not configured.
Extended Authentication not configured.
NAT-T is not detected
Cisco Trust Security SGT is disabled
Initiator of SA : Yes

IPv6 Crypto IKEv2 SA

R1#
```

show crypto ikev2 proposal

```
R1#show crypto ikev2 proposal
IKEv2 proposal: PROP-IKEV2
Encryption : AES-CBC-256
Integrity : SHA256
PRF : SHA256
DH Group : DH_GROUP_2048_MODP/Group 14
IKEv2 proposal: default
Encryption : AES-CBC-256 AES-CBC-192 AES-CBC-128
Integrity : SHA512 SHA384 SHA256 SHA96 MD596
PRF : SHA512 SHA384 SHA256 SHA1 MD5
DH Group : DH_GROUP_1536_MODP/Group 5 DH_GROUP_1024_MODP/Group 2

R1#
```

```
show crypto ikev2 profile
```

```
      Proposal : default
R1#show crypto ikev2 profile

IKEv2 profile: PROF-IKEV2
Ref Count: 2
Match criteria:
  Fvrf: global
  Local address/interface: none
  Identities:
    address 10.11.85.130 255.255.255.255
  Certificate maps: none
  Local identity: none
  Remote identity: none
  Local authentication method: pre-share
  Remote authentication method(s): pre-share
  EAP options: none
  Keyring: KR-IKEV2
  Trustpoint(s): none
  Lifetime: 86400 seconds
  DPD: disabled
  NAT-keepalive: disabled
  Ivrf: none
  Virtual-template: none
  AAA EAP authentication mlist: none
  AAA Accounting: none
  AAA group authorization: none
  AAA user authorization: none
R1#
```

```
! Access-list y crypto session
```

```
      Proposal : default
R2#sh access-list
Extended IP access list ACL-VPN-R2
  10 permit ip 10.11.85.192 0.0.0.63 10.11.85.0 0.0.0.63 (14 matches)
R2#
```

```
      10 permit ip 10.11.85.192 0.0.0.63 10.11.85.0 0.0.0.63 (14 matches)
R2#show crypto session
Crypto session current status

Interface: Ethernet0/0
Profile: PROF-IKEV2
Session status: UP-ACTIVE
Peer: 10.11.85.65 port 500
  Session ID: 1
  IKEV2 SA: local 10.11.85.130/500 remote 10.11.85.65/500 Active
  IPSEC FLOW: permit ip 10.11.85.192/255.255.255.192 10.11.85.0/255.255.255.192
    Active SAs: 2, origin: crypto map
R2#
```

